**Zen AI Model Family**

# Zen-Director

## Video Scene Generation, Storyboarding, and Cinematic Direction

**Technical Report v2025.01**

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January 2025

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**Abstract**

We present **Zen-Director**, a 7-billion parameter vision-language model specialized for video scene generation, cinematographic planning, and multi-shot storyboard synthesis. Built on the Zen MoDE (Mixture of Distilled Experts) architecture with a temporal transformer extension, Zen-Director understands and generates structured cinematic descriptions—shot lists, scene compositions, camera movements, and narrative arcs—that downstream video generation models can consume directly. The model achieves BLEU-4 of 0.423 on the VideoCaption benchmark, 81.2% accuracy on CinematicQA (our novel evaluation of cinematographic knowledge), and produces storyboards rated 4.1/5.0 by professional cinematographers in blind evaluation. Zen-Director bridges the gap between natural language creative intent and the precise technical specifications required to direct AI video generation systems at professional quality.

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# 1 Introduction

The rapid maturation of text-to-video generation systems has exposed a critical bottleneck: the gap between a creator's high-level creative intent and the precise technical specifications these systems require to produce coherent, professional-quality video. Current generation workflows demand that users specify camera angles, focal lengths, lighting conditions, shot durations, and scene transitions—knowledge traditionally held by professional cinematographers and directors.

Zen-Director addresses this bottleneck as a **directorial intelligence**: a model that understands narrative intent and translates it into actionable cinematic specifications. Rather than generating video pixels directly, Zen-Director generates structured **shot plans**: hierarchical scene descriptions that video generation systems consume as structured prompts.

## 1.1 Model Overview

| Property | Value |
|---|---|
| Parameters | 7B |
| Architecture | Zen MoDE 7B + Temporal Transformer |
| Context Length | 32K tokens + 512 video frames |
| Visual Encoder | ViT-L/14 (336px) |
| Temporal Depth | 12 temporal transformer layers |
| Shot Plan Format | Structured JSON + prose description |
| Training Data | 8.2M scene-annotation pairs, 1.4M film scripts |

Table 1: Zen-Director Model Specifications

## 1.2 Key Capabilities

- **Scene storyboarding**: Convert natural language scene descriptions into complete shot-by-shot storyboards with camera specifications.
- **Shot composition**: Recommend and generate compositional guidelines (rule of thirds, leading lines, depth of field) for each shot.
- **Narrative arc planning**: Structure multi-scene video narratives with consistent pacing, tension arcs, and visual motifs.
- **Cinematic vocabulary**: Understand and generate industry-standard cinematographic terminology (establishing shot, dolly zoom, rack focus, etc.).
- **Video comprehension**: Analyze existing video clips and generate directorial notes describing their cinematographic techniques.

# 2 Architecture

## 2.1 Zen MoDE 7B Language Backbone

The language backbone is Zen MoDE at 7B scale: 28 transformer layers, 28 attention heads with grouped-query attention (4 KV heads), and MoE feed-forward networks (4 experts, top-2 routing). This provides strong natural language understanding for parsing creative briefs and generating rich cinematic descriptions.

## 2.2 Visual Encoder

A ViT-L/14 vision encoder operating at 336px resolution encodes reference images and video frames into 256 visual tokens per frame. A two-layer MLP projection maps visual tokens into the language model's embedding space.

## 2.3 Temporal Transformer Extension

The key architectural innovation is a 12-layer temporal transformer that operates *across* frames rather than within them. Given $T$ frames, each encoded to 256 tokens, the temporal transformer attends over the time dimension to build a coherent representation of motion, continuity, and scene evolution:

$$H_{\text{temporal}} = \text{TransformerEncoder}([h_1^{CLS}, h_2^{CLS}, \ldots, h_T^{CLS}]) \tag{1}$$

where $h_t^{CLS}$ is the CLS token representation of frame $t$ from the visual encoder. The temporal representation is concatenated with text tokens before the language model's cross-attention layers.

## 2.4 Shot Plan Generation

Zen-Director generates shot plans as structured JSON objects:

```
{
  "scene": {
    "id": "s01",
    "location": "rain-soaked rooftop, night",
    "mood": "tense, noir",
    "duration_sec": 45
  },
  "shots": [
    {
      "id": "s01_001",
      "type": "establishing",
      "camera": {"position": "wide", "angle": "high angle", "move": "
          slow push in"},
      "focal_length": "24mm",
      "subject": "city skyline with protagonist silhouette",
      "duration_sec": 8,
      "lighting": "practical neon, rain reflections",
      "notes": "City should feel overwhelming relative to protagonist"
    }
  ]
}
```

Listing 1: Shot Plan Schema

# 3 Training

## 3.1 Dataset

| Source | Samples | Proportion | Content |
|---|---|---|---|
| Film scripts + frames | 2,400,000 | 29.3% | Script-to-scene alignment |
| Cinematography textbooks | 180,000 | 2.2% | Technical knowledge |
| Film criticism corpus | 820,000 | 10.0% | Aesthetic analysis |
| Video-caption pairs | 3,100,000 | 37.8% | Visual understanding |
| Storyboard collections | 1,700,000 | 20.7% | Shot plan examples |
| **Total** | **8,200,000** | 100% | |

Table 2: Zen-Director Training Data

## 3.2 Training Protocol

**Stage 1 – Visual encoder alignment** (20K steps): The ViT encoder and MLP projection are trained to align visual representations with cinematic language descriptions.

**Stage 2 – Temporal pretraining** (40K steps): The temporal transformer is pretrained on video sequences with a masked-frame prediction objective.

**Stage 3 – Directorial SFT** (60K steps): The full model is fine-tuned on storyboard generation, scene description, and cinematographic Q&A tasks jointly.

**Stage 4 – RLHF from cinematographers** (15K steps): A reward model trained on 50,000 pairwise comparisons from professional cinematographers (recruited via film school partnerships) is used to further refine shot plan quality via PPO.

# 4 Evaluation

## 4.1 VideoCaption Benchmark

We evaluate video description quality on a held-out set of 5,000 film clips spanning 20 genres.

| Model | BLEU-4 | METEOR | CIDEr | ROUGE-L |
|---|---|---|---|---|
| BLIP-2 | 0.281 | 0.314 | 0.872 | 0.512 |
| InstructBLIP | 0.308 | 0.341 | 0.941 | 0.534 |
| VideoChat2 | 0.352 | 0.374 | 1.021 | 0.561 |
| Video-LLaMA2 | 0.387 | 0.402 | 1.103 | 0.584 |
| **Zen-Director** | **0.423** | **0.441** | **1.187** | **0.612** |

Table 3: VideoCaption Benchmark Results (higher is better)

## 4.2 CinematicQA

CinematicQA is a novel benchmark we introduce comprising 2,000 multiple-choice questions testing cinematographic knowledge: shot types, camera movements, lighting techniques, editing principles, and genre conventions. Questions were authored by five professional cinematographers.

| Model | Accuracy (%) |
|---|---|
| GPT-4o (zero-shot) | 68.3 |
| Claude 3.5 Sonnet | 71.4 |
| Gemini 1.5 Pro | 66.8 |
| Specialist fine-tuned 7B | 74.2 |
| **Zen-Director 7B** | **81.2** |

Table 4: CinematicQA Accuracy (%)

### 4.3 Professional Storyboard Evaluation

Twenty professional cinematographers evaluated storyboards generated from 100 scene descriptions, rating each on a 5-point scale across five dimensions.

| Dimension | Mean Score (1–5) |
|---|---|
| Technical accuracy of shot specifications | 4.3 |
| Narrative coherence across shots | 4.1 |
| Creative quality / originality | 3.9 |
| Pacing appropriateness | 4.0 |
| Overall directorial vision | 4.1 |
| **Overall Mean** | **4.1** |

Table 5: Professional Cinematographer Evaluation (N=20 evaluators, 100 storyboards)

### 4.4 Downstream Video Generation Quality

We evaluate whether Zen-Director shot plans improve final video quality when used as structured prompts for a text-to-video generation system. Using the same creative brief:

| Prompt Method | Human Preference (%) | FVD ↓ |
|---|---|---|
| Raw creative brief (baseline) | 18.4% | 412 |
| Manual cinematographer spec | 42.1% | 287 |
| Zen-Director shot plan | 39.5% | 298 |

Table 6: Downstream Video Quality with Zen-Director Shot Plans

Zen-Director reaches 94% of manual cinematographer performance at a fraction of the cost and time, validating its role as an effective creative intermediary.

## 5 Applications

### 5.1 AI Film Production Pipeline

Zen-Director is integrated into the Hanzo AI film production pipeline as the directorial layer between a human creative brief and the Zen-Video generation system. A typical workflow:

1. Human writes a scene description in natural language.
2. Zen-Director generates a shot plan JSON with full cinematographic specs.
3. Human reviews and optionally edits the shot plan (typically 3–5 minutes).

4. Zen-Video consumes the shot plan and generates video clips per shot.
5. Zen-Director evaluates temporal consistency across clips and suggests retakes.

### 5.2 Film Education

Zen-Director serves as an interactive cinematography tutor: students can submit scene descriptions and receive expert-level directorial notes explaining the reasoning behind each shot choice.

### 5.3 Video Game Cutscene Direction

Game studios use Zen-Director to generate cinematic specifications for in-engine cutscene directors, reducing the time from narrative script to playable sequence by an estimated 60%.

## 6 Related Work

Video understanding models (VideoChat, Video-LLaMA) focus primarily on description generation. Text-to-video generation systems (Sora, CogVideo, ModelScope) focus on pixel synthesis. Zen-Director uniquely occupies the directorial planning layer between these two stages, drawing on the rich tradition of computational narrative research and cinematography theory.

## 7 Conclusion

Zen-Director establishes a new model category: the AI cinematographer. By training on film scripts, storyboards, and cinematographic knowledge at 7B scale, the model achieves 81.2% on CinematicQA and generates shot plans rated 4.1/5 by professional cinematographers. Integration with the Zen-Video generation backbone creates a complete AI film production pipeline from creative brief to rendered footage.

## References

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